Finite-Sample and Asymptotic Properties of the Linear GMM Estimator

Theory and Monte Carlo Evidence

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2026-08-31

Abstract

We give a self-contained treatment of the linear Generalized Method of Moments (GMM) estimator in an overidentified instrumental-variables model. We state the identification, consistency, and asymptotic-normality results from first principles, derive the efficiency bound attained by the optimal weighting matrix, and derive the limiting chi-squared distribution of Hansen's J statistic for testing overidentifying restrictions. We then contrast this asymptotic theory with finite-sample behaviour using a Monte Carlo experiment: an overidentified, conditionally heteroskedastic instrumental-variables design in which a one-step (2SLS-weighted) and a two-step efficient GMM estimator are compared across sample sizes ranging from 50 to 10,000. The simulation traces out consistency (bias and variance shrinking with n), the efficiency gain of two-step over one-step GMM, the approach of the studentized estimator's finite-sample distribution to its $N(0,1)$ limit, the coverage of sandwich-based confidence intervals, and the size of the J test, and discusses where each departs from its asymptotic approximation at small n.

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1 Introduction

The Generalized Method of Moments (GMM) estimator of Hansen (2022; see also Hayashi 2000; Hall 2005) is the workhorse estimator for overidentified linear models with endogenous regressors: it nests ordinary least squares, simple instrumental variables, and two-stage least squares (2SLS) as special cases, and it remains valid when the number of instruments exceeds the number of parameters and when the structural error is conditionally heteroskedastic. Its large-sample theory is standard textbook material – consistency, asymptotic normality, an efficiency bound attained by an optimal weighting matrix, and a chi-squared test of overidentifying restrictions. What that theory does *not* say is how large a sample has to be before its predictions become a good guide to what an applied researcher actually sees.

This handout has two purposes. Section 2 develops the finite-sample and asymptotic theory of the linear GMM estimator rigorously, from the moment condition through the closed-form estimator, its consistency and asymptotic normality, the efficiency bound, the two-step feasible efficient estimator, and Hansen's J test. Section 3 sets out a Monte Carlo design – an overidentified, conditionally heteroskedastic instrumental-variables model in which the one-step (2SLS-weighted) and two-step efficient GMM estimators genuinely differ – that is built to make each asymptotic result in Section 2 independently checkable in finite samples. Section 4 reports and interprets the simulation evidence, and Section 5 concludes.

2 Theoretical Background

2.1 The Linear Moment-Condition Model

Let $\{(y_i, \mathbf{x}_i, \mathbf{z}_i)\}_{i=1}^n$ be an independent and identically distributed sample, where y_i is a scalar outcome, $\mathbf{x}_i$ is a $K \times 1$ vector of regressors (possibly including endogenous variables), and $\mathbf{z}_i$ is an $L \times 1$ vector of instruments with $L \geq K$. The model is linear,

$$y_i = \mathbf{x}_i^\top \boldsymbol{\beta}_0 + \varepsilon_i,$$

and identification comes entirely from the population moment condition

$$\mathbb{E}[\mathbf{z}_i \varepsilon_i(\boldsymbol{\beta}_0)] = \mathbf{0}_L, \qquad \varepsilon_i(\boldsymbol{\beta}) \equiv y_i - \mathbf{x}_i^\top \boldsymbol{\beta}, \tag{1}$$

which states that the instruments are uncorrelated with the structural error at the true parameter value $\boldsymbol{\beta}_0$. When $L > K$, (1) imposes $L - K$ *overidentifying restrictions*: more moment conditions than are needed to pin down $\boldsymbol{\beta}_0$, which is what makes the specification testable (Section 2.6) and what makes an efficiency question ("which linear combination of the L conditions should we use?") meaningful in the first place.

Stacking observations into $\mathbf{X}$ $(n \times K)$, $\mathbf{Z}$ $(n \times L)$, and $\mathbf{y}$ $(n \times 1)$, define the *sample moment vector*

$$\bar{\mathbf{g}}_n(\boldsymbol{\beta}) \equiv \frac{1}{n}\mathbf{Z}^\top(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) = \frac{1}{n}\sum_{i=1}^n \mathbf{z}_i(y_i - \mathbf{x}_i^\top \boldsymbol{\beta}).$$

By the weak law of large numbers, $\bar{\mathbf{g}}_n(\boldsymbol{\beta}_0) \xrightarrow{p} \mathbb{E}[\mathbf{z}_i\varepsilon_i(\boldsymbol{\beta}_0)] = \mathbf{0}$, but for $L > K$ there is in general no $\boldsymbol{\beta}$ that sets $\bar{\mathbf{g}}_n(\boldsymbol{\beta})$ *exactly* to zero in a given sample: the L sample moment conditions are L equations in only K unknowns. GMM resolves this by choosing $\boldsymbol{\beta}$ to make the sample moments as close to zero as possible, in a quadratic metric.

2.2 The GMM Objective Function and the Closed-Form Estimator

For a symmetric, positive definite weighting matrix $\mathbf{W}_n$, the GMM estimator is

$$\hat{\boldsymbol{\beta}}(\mathbf{W}_n) = \arg\min_{\boldsymbol{\beta}} \; Q_n(\boldsymbol{\beta}), \qquad Q_n(\boldsymbol{\beta}) \equiv \bar{\mathbf{g}}_n(\boldsymbol{\beta})^\top \mathbf{W}_n \, \bar{\mathbf{g}}_n(\boldsymbol{\beta}). \tag{2}$$

Because the model is linear, Q_n is an exact quadratic form in $\boldsymbol{\beta}$ and (2) has a closed-form solution. Writing $\bar{\mathbf{g}}_n(\boldsymbol{\beta}) = \frac{1}{n}\mathbf{Z}^\top\mathbf{y} - \frac{1}{n}\mathbf{Z}^\top\mathbf{X}\boldsymbol{\beta}$, the first-order condition $\partial Q_n/\partial\boldsymbol{\beta} = -\frac{2}{n}\mathbf{X}^\top\mathbf{Z}\,\mathbf{W}_n\,\bar{\mathbf{g}}_n(\boldsymbol{\beta}) = \mathbf{0}$ gives

$$\mathbf{X}^\top\mathbf{Z}\,\mathbf{W}_n\,\mathbf{Z}^\top\mathbf{X}\,\hat{\boldsymbol{\beta}}(\mathbf{W}_n) = \mathbf{X}^\top\mathbf{Z}\,\mathbf{W}_n\,\mathbf{Z}^\top\mathbf{y},$$

so that, whenever $\mathbf{X}^\top\mathbf{Z}\,\mathbf{W}_n\,\mathbf{Z}^\top\mathbf{X}$ is invertible,

$$\hat{\boldsymbol{\beta}}(\mathbf{W}_n) = (\mathbf{X}^\top\mathbf{Z}\,\mathbf{W}_n\,\mathbf{Z}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{Z}\,\mathbf{W}_n\,\mathbf{Z}^\top\mathbf{y}. \tag{3}$$

Two familiar estimators are special cases of (3): $\mathbf{W}_n = \mathbf{I}_L$ gives unweighted GMM, and $\mathbf{W}_n = (\mathbf{Z}^\top\mathbf{Z}/n)^{-1}$ gives numerically the 2SLS estimator. Neither is asymptotically efficient in general; Section 2.5 derives the weighting matrix that is.

2.3 Consistency

Assumption 1 (Identification and Regularity)**.**

1. **(Correct specification)** $\mathbb{E}[\mathbf{z}_i\varepsilon_i(\boldsymbol{\beta})] = \mathbf{0}$ holds if and only if $\boldsymbol{\beta} = \boldsymbol{\beta}_0$, for $\boldsymbol{\beta}_0$ in the interior of a compact parameter space.
2. **(Rank condition)** $\mathbf{G} \equiv \mathbb{E}[\mathbf{z}_i\mathbf{x}_i^\top]$ has full column rank K (in particular $L \geq K$).
3. **(Weighting matrix)** $\mathbf{W}_n \xrightarrow{p} \mathbf{W}$, a finite, symmetric, positive definite $L \times L$ matrix.
4. **(Moments)** $\mathbb{E}\|\mathbf{z}_i\mathbf{x}_i^\top\| < \infty$ and $\mathbb{E}\|\mathbf{z}_i y_i\| < \infty$.

Proposition 1 (Consistency of Linear GMM)**.** *Under Assumption 1,* $\hat{\boldsymbol{\beta}}(\mathbf{W}_n) \xrightarrow{p} \boldsymbol{\beta}_0$.

Proof. Substituting $y_i = \mathbf{x}_i^\top\boldsymbol{\beta}_0 + \varepsilon_i$ into $\bar{\mathbf{g}}_n(\boldsymbol{\beta}_0) = \frac{1}{n}\mathbf{Z}^\top\boldsymbol{\varepsilon}$ and writing $\hat{\mathbf{G}}_n \equiv \frac{1}{n}\mathbf{Z}^\top\mathbf{X}$, the closed form (3) rearranges to

$$\hat{\boldsymbol{\beta}}(\mathbf{W}_n) - \boldsymbol{\beta}_0 = (\hat{\mathbf{G}}_n^\top\mathbf{W}_n\hat{\mathbf{G}}_n)^{-1}\hat{\mathbf{G}}_n^\top\mathbf{W}_n\,\bar{\mathbf{g}}_n(\boldsymbol{\beta}_0). \tag{4}$$

By the weak law of large numbers, $\hat{\mathbf{G}}_n \xrightarrow{p} \mathbf{G}$ and $\bar{\mathbf{g}}_n(\boldsymbol{\beta}_0) \xrightarrow{p} \mathbb{E}[\mathbf{z}_i\varepsilon_i(\boldsymbol{\beta}_0)] = \mathbf{0}$ by Assumption 1.1. Combined with $\mathbf{W}_n \xrightarrow{p} \mathbf{W}$ and the continuous-mapping theorem, $\mathbf{G}^\top\mathbf{W}\mathbf{G}$ is invertible by Assumption 1.2–1.3 (full column rank $\mathbf{G}$ and positive definite $\mathbf{W}$ imply $\mathbf{G}^\top\mathbf{W}\mathbf{G}$ is positive definite, hence invertible), so the right-hand side of (4) converges in probability to $(\mathbf{G}^\top\mathbf{W}\mathbf{G})^{-1}\mathbf{G}^\top\mathbf{W}\cdot\mathbf{0} = \mathbf{0}$. ■ ■

Two features of this argument are worth flagging because the simulation is built around them. First, consistency holds for *every* positive definite limiting weight $\mathbf{W}$ – the one-step and two-step estimators of Section 2.5 are both consistent, so any efficiency difference between them is a finite-sample or asymptotic-variance phenomenon, not a difference in the target they estimate. Second, the proof uses only a law of large numbers; no distributional assumption on ε_i, and in particular no homoskedasticity, is needed.

2.4 Asymptotic Normality

Assumption 2 (Central Limit Theorem for the Moments)**.** In addition to Assumption 1, $\mathbf{S} \equiv \operatorname{Var}(\mathbf{z}_i \varepsilon_i(\boldsymbol{\beta}_0)) = \mathbb{E}[\mathbf{z}_i\mathbf{z}_i^\top \varepsilon_i^2]$ is finite and positive definite, and

$$\sqrt{n}\,\bar{\mathbf{g}}_n(\boldsymbol{\beta}_0) = \frac{1}{\sqrt{n}}\mathbf{Z}^\top \boldsymbol{\varepsilon} \xrightarrow{d} N(\mathbf{0}, \mathbf{S}).$$

Assumption 2 allows ε_i to be conditionally heteroskedastic – $\mathbb{E}[\varepsilon_i^2 \mid \mathbf{z}_i]$ need not be constant – which is exactly the feature built into the Monte Carlo design below, and is why $\mathbf{S}$ is written as $\mathbb{E}[\mathbf{z}_i\mathbf{z}_i^\top \varepsilon_i^2]$ rather than $\sigma^2\mathbb{E}[\mathbf{z}_i\mathbf{z}_i^\top]$.

Proposition 2 (Asymptotic Normality of Linear GMM)**.** *Under Assumptions 1–2,*

$$\sqrt{n}(\hat{\boldsymbol{\beta}}(\mathbf{W}_n) - \boldsymbol{\beta}_0) \xrightarrow{d} N\big(\mathbf{0},\ \mathbf{V}(\mathbf{W})\big), \qquad \mathbf{V}(\mathbf{W}) = (\mathbf{G}^\top\mathbf{W}\mathbf{G})^{-1}\mathbf{G}^\top\mathbf{W}\mathbf{S}\mathbf{W}\mathbf{G}(\mathbf{G}^\top\mathbf{W}\mathbf{G})^{-1}. \tag{5}$$

Proof. Multiply (4) by $\sqrt{n}$:

$$\sqrt{n}(\hat{\boldsymbol{\beta}}(\mathbf{W}_n) - \boldsymbol{\beta}_0) = (\hat{\mathbf{G}}_n^\top\mathbf{W}_n\hat{\mathbf{G}}_n)^{-1}\hat{\mathbf{G}}_n^\top\mathbf{W}_n\ \sqrt{n}\,\bar{\mathbf{g}}_n(\boldsymbol{\beta}_0).$$

By Assumption 2, $\sqrt{n}\,\bar{\mathbf{g}}_n(\boldsymbol{\beta}_0) \xrightarrow{d} N(\mathbf{0}, \mathbf{S})$; by Assumption 1, $\hat{\mathbf{G}}_n \xrightarrow{p} \mathbf{G}$ and $\mathbf{W}_n \xrightarrow{p} \mathbf{W}$. Slutsky's theorem applied to the product of the (stochastically bounded, converging-in-probability) matrix $(\hat{\mathbf{G}}_n^\top\mathbf{W}_n\hat{\mathbf{G}}_n)^{-1}\hat{\mathbf{G}}_n^\top\mathbf{W}_n$ and the asymptotically normal vector $\sqrt{n}\,\bar{\mathbf{g}}_n(\boldsymbol{\beta}_0)$ gives

$$\sqrt{n}(\hat{\boldsymbol{\beta}}(\mathbf{W}_n) - \boldsymbol{\beta}_0) \xrightarrow{d} (\mathbf{G}^\top\mathbf{W}\mathbf{G})^{-1}\mathbf{G}^\top\mathbf{W}\,\mathbf{u}, \qquad \mathbf{u} \sim N(\mathbf{0}, \mathbf{S}),$$

and the variance of a linear transformation $\mathbf{A}\mathbf{u}$ of $\mathbf{u} \sim N(\mathbf{0}, \mathbf{S})$ is $\mathbf{A}\mathbf{S}\mathbf{A}^\top$, which for $\mathbf{A} = (\mathbf{G}^\top\mathbf{W}\mathbf{G})^{-1}\mathbf{G}^\top\mathbf{W}$ gives exactly (5). ■ ■

In a finite sample, (5) is implemented by plugging in consistent estimators $\hat{\mathbf{G}}_n = \frac{1}{n}\mathbf{Z}^\top\mathbf{X}$, $\hat{\mathbf{S}}_n$ (Section 2.5), and $\mathbf{W}_n$, giving the *sandwich* variance estimator

$$\widehat{\operatorname{Var}}(\hat{\boldsymbol{\beta}}(\mathbf{W}_n)) = \frac{1}{n}(\hat{\mathbf{G}}_n^\top\mathbf{W}_n\hat{\mathbf{G}}_n)^{-1}\hat{\mathbf{G}}_n^\top\mathbf{W}_n\hat{\mathbf{S}}_n\mathbf{W}_n\hat{\mathbf{G}}_n(\hat{\mathbf{G}}_n^\top\mathbf{W}_n\hat{\mathbf{G}}_n)^{-1}. \tag{6}$$

This is the standard-error formula used for both estimators in the simulation; it collapses to a simpler expression for the efficient estimator below.

2.5 Efficient GMM and the Two-Step Estimator

Equation (5) shows that the asymptotic variance depends on the choice of $\mathbf{W}$. Hansen (2022)'s classical result identifies the efficient choice.

Proposition 3 (GMM Efficiency Bound)**.** *Among all weighting matrices* $\mathbf{W}$ *satisfying Assumption 1.3,* $\mathbf{V}(\mathbf{W}) - \mathbf{V}(\mathbf{S}^{-1})$ *is positive semi-definite for every* $\mathbf{W}$*, so* $\mathbf{W} = \mathbf{S}^{-1}$ *minimizes the asymptotic variance (5) in the positive semi-definite ordering, and*

$$\mathbf{V}(\mathbf{S}^{-1}) = (\mathbf{G}^\top\mathbf{S}^{-1}\mathbf{G})^{-1}. \tag{7}$$

Proof. Substituting $\mathbf{W} = \mathbf{S}^{-1}$ into (5) gives $\mathbf{V}(\mathbf{S}^{-1}) = (\mathbf{G}^\top\mathbf{S}^{-1}\mathbf{G})^{-1}\mathbf{G}^\top\mathbf{S}^{-1}\mathbf{S}\mathbf{S}^{-1}\mathbf{G}(\mathbf{G}^\top\mathbf{S}^{-1}\mathbf{G})^{-1} = (\mathbf{G}^\top\mathbf{S}^{-1}\mathbf{G})^{-1}$, which is (7). For optimality, let $\mathbf{S}^{1/2}$ be a symmetric square root of $\mathbf{S}$ and define $\mathbf{A} \equiv (\mathbf{G}^\top\mathbf{W}\mathbf{G})^{-1}\mathbf{G}^\top\mathbf{W}\mathbf{S}^{1/2}$ and $\mathbf{B} \equiv (\mathbf{G}^\top\mathbf{S}^{-1}\mathbf{G})^{-1}\mathbf{G}^\top\mathbf{S}^{-1/2}$, so that $\mathbf{V}(\mathbf{W}) = \mathbf{A}\mathbf{A}^\top$ and $\mathbf{V}(\mathbf{S}^{-1}) = \mathbf{B}\mathbf{B}^\top$. Direct multiplication using $\mathbf{S}^{-1/2}\mathbf{S}^{1/2} = \mathbf{I}_L$ verifies $\mathbf{B}\mathbf{A}^\top = \mathbf{B}\mathbf{B}^\top$, i.e. $(\mathbf{A} - \mathbf{B})\mathbf{B}^\top = \mathbf{0}$, so

$\mathbf{A}\mathbf{A}^\top = \mathbf{B}\mathbf{B}^\top + (\mathbf{A}-\mathbf{B})(\mathbf{A}-\mathbf{B})^\top$, and $(\mathbf{A}-\mathbf{B})(\mathbf{A}-\mathbf{B})^\top$ is positive semi-definite. Hence $\mathbf{V}(\mathbf{W}) - \mathbf{V}(\mathbf{S}^{-1}) = (\mathbf{A}-\mathbf{B})(\mathbf{A}-\mathbf{B})^\top \geq \mathbf{0}$. ■ ■

The efficiency bound (7) is attractive but infeasible: $\mathbf{S}$ is unknown. The standard solution is **two-step efficient GMM**:

1. **Step 1.** Choose any consistent, feasible weighting matrix – in this handout, $\mathbf{W}_n^{(1)} = (\mathbf{Z}^\top\mathbf{Z}/n)^{-1}$, which makes $\hat{\boldsymbol{\beta}}^{(1)} \equiv \hat{\boldsymbol{\beta}}(\mathbf{W}_n^{(1)})$ numerically equal to the 2SLS estimator. By Proposition 1, $\hat{\boldsymbol{\beta}}^{(1)}$ is consistent even though $\mathbf{W}_n^{(1)}$ is generally *not* efficient.
2. **Estimate S.** Using the step-1 residuals $\hat{\varepsilon}_i^{(1)} = y_i - \mathbf{x}_i^\top\hat{\boldsymbol{\beta}}^{(1)}$, form the heteroskedasticity-robust estimator
$$\hat{\mathbf{S}}_n = \frac{1}{n}\sum_{i=1}^{n}\mathbf{z}_i\mathbf{z}_i^\top(\hat{\varepsilon}_i^{(1)})^2 \xrightarrow{p} \mathbf{S},$$
which is consistent for $\mathbf{S}$ under Assumption 2 regardless of the form of the conditional heteroskedasticity (White 2000).
3. **Step 2.** Set $\mathbf{W}_n^{(2)} = \hat{\mathbf{S}}_n^{-1}$ and compute $\hat{\boldsymbol{\beta}}^{(2)} \equiv \hat{\boldsymbol{\beta}}(\mathbf{W}_n^{(2)})$.

Because $\mathbf{W}_n^{(2)} \xrightarrow{p} \mathbf{S}^{-1}$, Proposition 3 applies with $\mathbf{W} = \mathbf{S}^{-1}$, so $\sqrt{n}(\hat{\boldsymbol{\beta}}^{(2)} - \boldsymbol{\beta}_0) \xrightarrow{d} N(\mathbf{0}, (\mathbf{G}^\top\mathbf{S}^{-1}\mathbf{G})^{-1})$, and the sandwich formula (6) collapses to

$$\widehat{\mathrm{Var}}(\hat{\boldsymbol{\beta}}^{(2)}) = \frac{1}{n}(\hat{\mathbf{G}}_n^\top\hat{\mathbf{S}}_n^{-1}\hat{\mathbf{G}}_n)^{-1}. \tag{8}$$

This is the estimator the simulation calls the "two-step efficient GMM" estimator, and (8) is the variance used for its confidence intervals.

2.6 Testing Overidentifying Restrictions

When $L > K$, correct specification (Assumption 1.1) is testable: it implies $\bar{\mathbf{g}}_n(\hat{\boldsymbol{\beta}}^{(2)})$ should be close to zero even though nothing in the estimation *forces* all L sample moments to vanish. Hansen's J statistic formalizes "close to zero."

Proposition 4 (Hansen's J Test)**.** *Under Assumptions 1–2 and correct specification,*

$$J_n \equiv n\,\bar{\mathbf{g}}_n(\hat{\boldsymbol{\beta}}^{(2)})^\top\hat{\mathbf{S}}_n^{-1}\bar{\mathbf{g}}_n(\hat{\boldsymbol{\beta}}^{(2)}) \xrightarrow{d} \chi^2(L-K).$$

Proof. A first-order Taylor expansion of $\sqrt{n}\bar{\mathbf{g}}_n(\hat{\boldsymbol{\beta}}^{(2)})$ around $\boldsymbol{\beta}_0$, combined with the first-order condition defining $\hat{\boldsymbol{\beta}}^{(2)}$, gives

$$\sqrt{n}\,\bar{\mathbf{g}}_n(\hat{\boldsymbol{\beta}}^{(2)}) = \left[\mathbf{I}_L - \mathbf{G}(\mathbf{G}^\top\mathbf{S}^{-1}\mathbf{G})^{-1}\mathbf{G}^\top\mathbf{S}^{-1}\right]\sqrt{n}\,\bar{\mathbf{g}}_n(\boldsymbol{\beta}_0) + o_p(1).$$

Premultiplying by $\mathbf{S}^{-1/2}$ and writing $\mathbf{u} \equiv \mathbf{S}^{-1/2}\sqrt{n}\,\bar{\mathbf{g}}_n(\boldsymbol{\beta}_0) \xrightarrow{d} N(\mathbf{0}, \mathbf{I}_L)$ (Assumption 2) and $\mathbf{P} \equiv \mathbf{S}^{-1/2}\mathbf{G}(\mathbf{G}^\top\mathbf{S}^{-1}\mathbf{G})^{-1}\mathbf{G}^\top\mathbf{S}^{-1/2}$, this is

$$\mathbf{S}^{-1/2}\sqrt{n}\,\bar{\mathbf{g}}_n(\hat{\boldsymbol{\beta}}^{(2)}) = (\mathbf{I}_L - \mathbf{P})\,\mathbf{u} + o_p(1).$$

$\mathbf{P}$ is idempotent ($\mathbf{P}^2 = \mathbf{P}$, by direct multiplication) and symmetric, with $\mathrm{rank}(\mathbf{P}) = \mathrm{tr}(\mathbf{P}) = \mathrm{tr}((\mathbf{G}^\top\mathbf{S}^{-1}\mathbf{G})^{-1}\mathbf{G}^\top\mathbf{S}^{-1}\mathbf{G}) = \mathrm{tr}(\mathbf{I}_K) = K$, so $\mathbf{I}_L - \mathbf{P}$ is itself an idempotent projection of rank $L-K$. Since $J_n \xrightarrow{d} \mathbf{u}^\top(\mathbf{I}_L - \mathbf{P})\mathbf{u}$ and $\mathbf{u} \sim N(\mathbf{0}, \mathbf{I}_L)$, this is the quadratic form of a standard normal vector in an idempotent matrix of rank $L-K$, which is $\chi^2(L-K)$ by the standard result for quadratic forms in normal vectors. ■ ■

With one endogenous regressor and two excluded instruments (Section 3), the design below has $L = 3$, $K = 2$, and a single overidentifying restriction, so $J_n \xrightarrow{d} \chi^2(1)$ under correct specification.

2.7 Finite-Sample Properties: What Asymptotic Theory Does Not Guarantee

Propositions 1–4 are exact statements about limits as $n \to \infty$; none of them bounds how large n must be in practice, and each can fail badly in a finite sample that its own asymptotic approximation does not warn against. Three caveats motivate the simulation design in Section 3:

1. **GMM estimators are, in general, biased in finite samples**, even though they are consistent. Unlike OLS under the classical Gauss–Markov assumptions, $\mathbb{E}[\hat{\boldsymbol{\beta}}(\mathbf{W}_n)] \neq \boldsymbol{\beta}_0$ at any fixed n for the linear IV/GMM estimator, because $\hat{\boldsymbol{\beta}}(\mathbf{W}_n)$ is a nonlinear (ratio-like) function of the sample moments even though the *population* moment condition is linear in $\boldsymbol{\beta}$; the bias is $O(1/n)$ under standard regularity conditions and shrinks as identification strengthens (Hall 2005, ch. 3; Hayashi 2000, sec. 3.4).
2. **Two-step efficient GMM introduces an extra source of finite-sample noise**: $\hat{\mathbf{S}}_n$ in Section 2.5 is itself estimated, using step-1 residuals, and that estimation error propagates into $\hat{\boldsymbol{\beta}}^{(2)}$. In sufficiently small samples, or with many overidentifying moments, this can erode or even reverse the asymptotic efficiency ranking of Proposition 3 relative to a well-chosen one-step estimator – a concern that motivates alternatives such as continuously-updated GMM (Hall 2005, ch. 4).
3. **Weak or many instruments distort both estimation and inference.** When $\mathbf{G}$ is close to reduced rank (weak identification) the finite-sample distribution of $\hat{\boldsymbol{\beta}}(\mathbf{W}_n)$ can be far from normal and biased toward the OLS estimator's plim, and the sandwich standard errors (6) and (8) can be badly miscalibrated even at moderate n (Chaussé 2025).

The design in Section 3 deliberately avoids weak identification (caveat 3) so that the simulation isolates caveats 1–2 – ordinary finite-sample bias and the cost of estimating the optimal weighting matrix – against the backdrop of Propositions 1–4.

3 Monte Carlo Design

3.1 Data-Generating Process

The design has one endogenous regressor and two excluded, exogenous instruments, so $L = 3$ instruments $(1, z_{1i}, z_{2i})$ and $K = 2$ regressors $(1, x_{2i})$: one overidentifying restriction. For each replication:

$$z_{1i}, z_{2i} \overset{\text{iid}}{\sim} N(0,1), \qquad u_i, w_i \overset{\text{iid}}{\sim} N(0,1), \quad u_i, w_i \perp z_{1i}, z_{2i},$$

$$x_{2i} = \pi_0 + \pi_1 z_{1i} + \pi_2 z_{2i} + u_i, \qquad e_i = \rho\, u_i + \sqrt{1-\rho^2}\, w_i, \qquad \varepsilon_i = \sqrt{0.5 + z_{1i}^2}\, e_i,$$

$$y_i = \beta_1 + \beta_2 x_{2i} + \varepsilon_i.$$

The first-stage error u_i enters both x_{2i} and (through e_i) the structural error ε_i, so x_{2i} is endogenous with $\mathrm{Corr}(u_i, e_i) = \rho$; the parametrization $e_i = \rho u_i + \sqrt{1-\rho^2} w_i$ with u_i, w_i independent standard normal makes this exact regardless of $\rho \in (-1, 1)$. The multiplicative factor $\sqrt{0.5 + z_{1i}^2}$ makes $\mathrm{Var}(\varepsilon_i \mid z_{1i}, z_{2i}) = 0.5 + z_{1i}^2$ depend on the instrument z_{1i}: the errors are conditionally heteroskedastic, so $\mathbf{S} = \mathbb{E}[\mathbf{z}_i \mathbf{z}_i^\top \varepsilon_i^2] \neq \sigma^2 \mathbb{E}[\mathbf{z}_i \mathbf{z}_i^\top]$ and the one-step (2SLS-weighted) and two-step efficient GMM estimators of Section 2.5 have genuinely different asymptotic variances – without heteroskedasticity the two would coincide asymptotically and the comparison in Section 4.1 would be uninformative.

Parameter values, held fixed across all replications and sample sizes:

$$(\beta_1, \beta_2) = (1, 2), \qquad (\pi_0, \pi_1, \pi_2) = (0.5, 1, 1), \qquad \rho = 0.6.$$

The instruments are relevant ($\pi_1 = \pi_2 = 1$, not weak) and the endogeneity is moderate ($\rho = 0.6$), consistent with the decision in Section 2.6 to isolate ordinary finite-sample bias and weighting-matrix estimation error rather than weak-instrument pathologies.

3.2 Estimators Compared

For each simulated sample, two estimators are computed from the identical data, exactly as in Section 2.5:

- **One-step GMM** ($\hat{\beta}^{(1)}$): $\mathbf{W}_n^{(1)} = (\mathbf{Z}^\top \mathbf{Z}/n)^{-1}$, numerically identical to 2SLS, with sandwich standard errors from (6).
- **Two-step efficient GMM** ($\hat{\beta}^{(2)}$): $\mathbf{W}_n^{(2)} = \hat{\mathbf{S}}_n^{-1}$, with $\hat{\mathbf{S}}_n$ the heteroskedasticity-robust estimator built from the $\hat{\beta}^{(1)}$ residuals, and standard errors from the simplified formula (8).

Both estimators are consistent for $\boldsymbol{\beta}_0$ by Proposition 1; only their asymptotic (and finite-sample) variance should differ. Hansen's J statistic of Section 2.6 is computed at $\hat{\beta}^{(2)}$ using $\mathbf{W}_n^{(2)}$, with $L - K = 1$ degree of freedom.

3.3 Simulation Parameters and Reported Statistics

The sample size grid is $n \in \{50, 100, 250, 500, 1000, 2500, 5000, 10000\}$, with $N_{sim} = 5000$ independent replications drawn at each n (seed fixed at the start of the experiment for reproducibility). All reported quantities concern the slope β_2, the coefficient on the endogenous regressor; the intercept behaves analogously and is omitted. At each n the script `gmm_finite_sample_asymptotic_sim.R` computes, across the N_{sim} replications:

- **Bias** and **variance** of $\hat{\beta}_2^{(1)}$ and $\hat{\beta}_2^{(2)}$ around the true β_2 (Section 4.1: consistency and efficiency);
- **RMSE** of both estimators;
- **Coverage** of the nominal 95% confidence interval $\hat{\beta}_2 \pm 1.96 \times \widehat{\mathrm{se}}(\hat{\beta}_2)$, using the sandwich standard errors of (6) and (8) (Section 4.3);
- The rejection frequency of Hansen's J_n statistic at the nominal 5% level against $\chi^2(1)$ (Section 4.4).

In addition, at a small sample size ($n = 50$) and a large one ($n = 5000$), the script retains every replication's studentized statistic $(\hat{\beta}_2^{(2)} - \beta_2)/\widehat{\mathrm{se}}(\hat{\beta}_2^{(2)})$, which Proposition 2 predicts converges to $N(0, 1)$, to visualize how close that limit is at each n (Section 4.2).

4 Results

4.1 Consistency and Efficiency

Figure 1 plots Monte Carlo bias against n (log scale) for both estimators. Both are visually indistinguishable from zero once n reaches a few hundred, and neither series shows a systematic sign or trend as n grows further – exactly what Proposition 1 predicts, since both weighting matrices satisfy Assumption 1.3 regardless of their efficiency. At the smallest sample size, $n = 50$, the one-step estimator has bias -0.0029 and the two-step estimator -0.0044; by $n = 10^4$ both have shrunk an order of magnitude further, to 0.0002 and 0.0002 respectively – consistent with the $O(1/n)$ finite-sample bias flagged in Section 2.7, caveat 1, which predicts bias shrinking in proportion to n, faster than the $O(1/\sqrt{n})$ rate of the standard error itself.

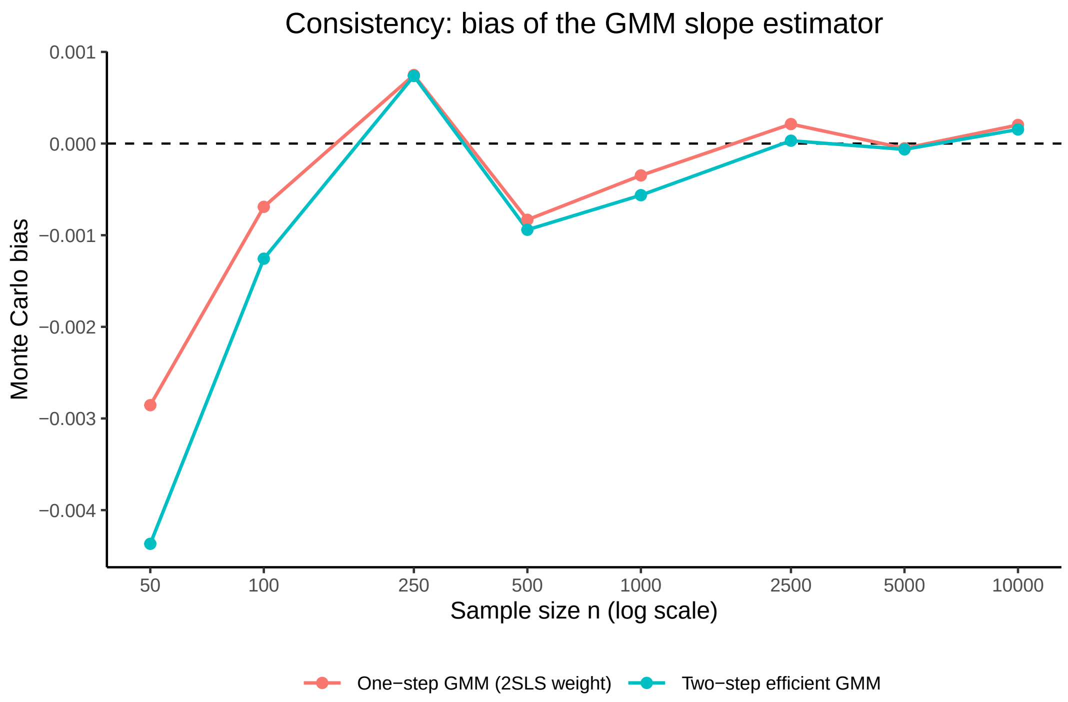

Figure 1: Monte Carlo bias of the one-step (2SLS-weighted) and two-step efficient GMM slope estimators, by sample size.

Figure 2 plots Monte Carlo variance against n on a log-log scale. Two features match Section 2 exactly. First, both curves are close to a straight line of slope -1, the $O(1/n)$ rate implied by (5)/(7) (variance $\approx \mathbf{V}/n$ for fixed asymptotic $\mathbf{V}$). Second, the two-step efficient estimator's variance lies *below* the one-step estimator's at **every** sample size in the grid – the efficiency ranking of Proposition 3 is not just an asymptotic statement here but holds throughout the range simulated. The efficiency gain, $1 - \widehat{\text{Var}}(\hat{\beta}^{(2)})/\widehat{\text{Var}}(\hat{\beta}^{(1)})$, is 12.0% at $n = 50$ and 17.8% at $n = 10^4$: the efficient estimator's advantage, driven by the conditional heteroskedasticity built into the design (Section 3.1), is present even in small samples and if anything widens as n grows and $\hat{\mathbf{S}}_n$ is estimated more precisely.

4.2 Asymptotic Normality

Proposition 2 predicts that the studentized statistic $(\hat{\beta}_2^{(2)} - \beta_2)/\widehat{\text{se}}(\hat{\beta}_2^{(2)})$ is approximately $N(0, 1)$, with the approximation improving as $n \to \infty$. Figure 3 overlays the Monte Carlo density of this statistic against the $N(0, 1)$ density at $n = 50$ and $n = 5000$, and Figure 4 shows the corresponding QQ-plots against theoretical normal quantiles. At $n = 50$ the histogram is visibly heavier in the centre and the QQ-plot bows away from the 45-degree line in the tails – a finite-sample departure from normality consistent with the $O(1/n)$ bias and imperfectly estimated $\hat{\mathbf{S}}_n$ discussed in Section 2.7. At $n = 5000$ both diagnostics track the normal reference closely, illustrating Proposition 2's limit being approached, not merely asserted.

4.3 Coverage of Confidence Intervals

Figure 5 plots the empirical coverage of the nominal 95% confidence interval, built from the sandwich standard errors (6) and (8), against n. At $n = 50$ coverage is 91.8% for the one-step estimator and 91.4% for the two-step estimator – both below the nominal 95%, the direct consequence of the finite-sample bias and non-normality documented in Sections 4.1–4.2 feeding into an interval that assumes exact

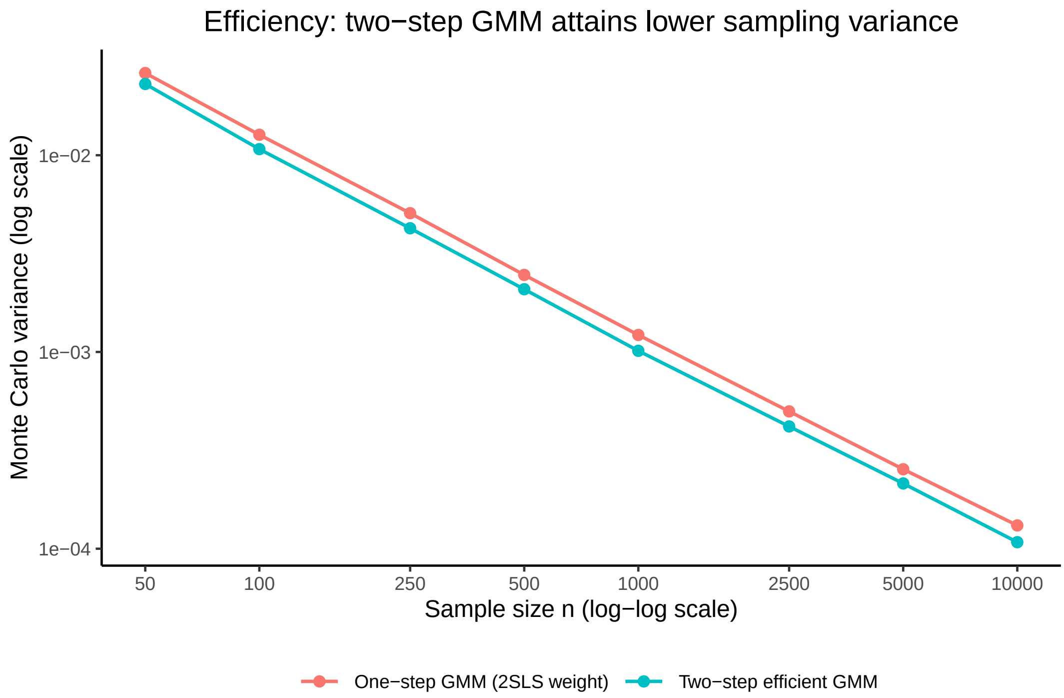

Figure 2: Monte Carlo variance of the two GMM slope estimators, log-log scale; the two-step efficient estimator (Proposition 3) attains lower variance at every sample size.

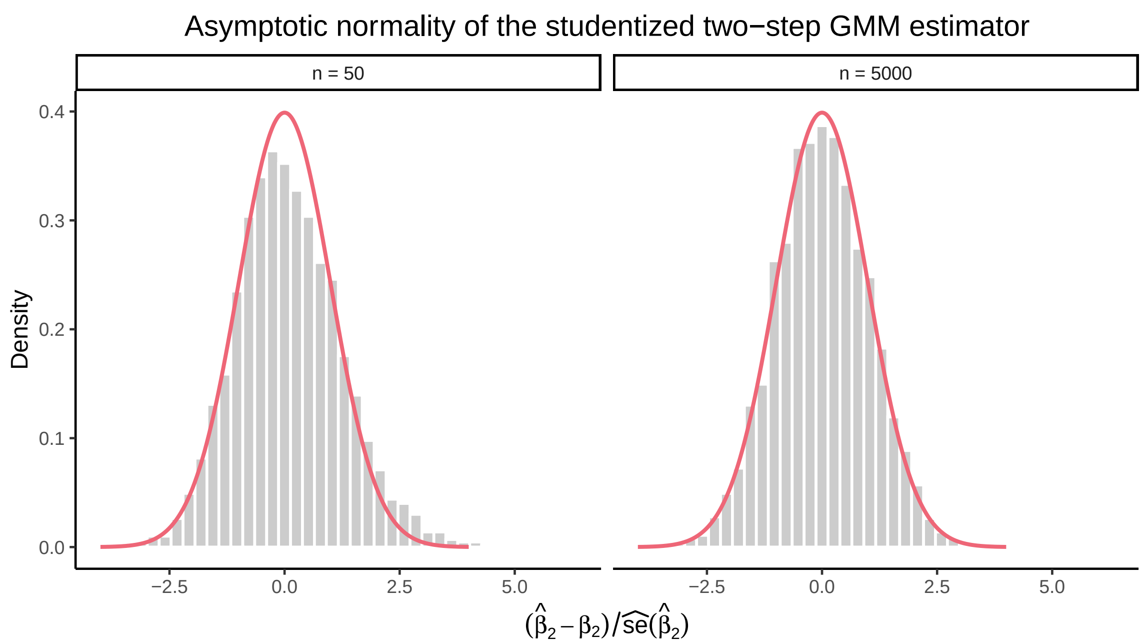

Figure 3: Density of the studentized two-step GMM slope estimator at a small and a large sample size, against its N(0,1) asymptotic limit.

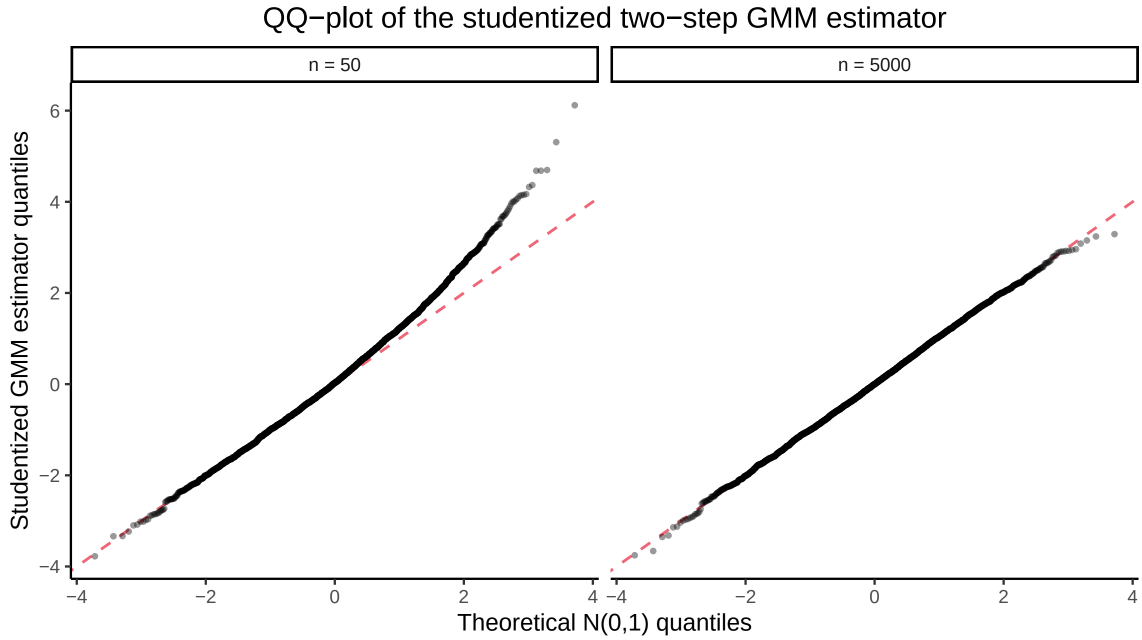

Figure 4: QQ-plot of the studentized two-step GMM slope estimator against theoretical normal quantiles.

normality and a correctly centred point estimate. Coverage rises toward the nominal level as n grows, reaching 94.5% and 94.8% respectively at $n = 10^4$, consistent with Proposition 2. This is the practical stakes of Section 2.7: a researcher quoting a 95% CI from GMM estimated on a few dozen observations is, in this design, actually covering the true slope somewhat less often than the label promises.

4.4 Size of the Overidentification Test

Figure 6 plots the rejection frequency of Hansen's J_n statistic at the nominal 5% level against $\chi^2(1)$, as a function of n. Because the model is correctly specified at every n (the same DGP generates the data used to estimate and to test it), Proposition 4 predicts a rejection frequency of exactly 5% in the limit; this is a test of **size**, not power. The rejection frequency is 5.3% at $n = 50$ and settles near the nominal 5.0% for $n \geq 250$, confirming that the $\chi^2(1)$ approximation of Proposition 4 is reliable well before the sample sizes at which the coverage results of Section 4.3 fully converge – the J test's asymptotic approximation kicks in faster here than the confidence-interval coverage does, because the J statistic depends on $\hat{\mathbf{S}}_n$ only through its role in weighting, whereas coverage also depends on the point estimate's own finite-sample bias.

4.5 Summary Table

Table 1 collects all of the reported statistics by sample size.

Table 1: Monte Carlo bias, variance, RMSE, 95% CI coverage, and J-test rejection frequency of the one-step and two-step GMM slope estimators (N_sim = 5000 replications per row).

n	Bias (1-step)	Bias (2-step)	Var (1-step)	Var (2-step)	RMSE (1-step)	RMSE (2-step)
50	-0.0029	-0.0044	0.02616	0.02302	0.1618	0.1518
100	-0.0007	-0.0013	0.01271	0.01074	0.1127	0.1036

Table 1: Monte Carlo bias, variance, RMSE, 95% CI coverage, and J-test rejection frequency of the one-step and two-step GMM slope estimators (N_sim = 5000 replications per row).

n	Bias (1-step)	Bias (2-step)	Var (1-step)	Var (2-step)	RMSE (1-step)	RMSE (2-step)
250	0.0007	0.0007	0.00507	0.00425	0.0712	0.0652
500	-0.0008	-0.0009	0.00246	0.00208	0.0496	0.0457
1000	-0.0003	-0.0006	0.00122	0.00101	0.0349	0.0318
2500	0.0002	0.0000	0.00050	0.00042	0.0223	0.0204
5000	-0.0001	-0.0001	0.00025	0.00021	0.0159	0.0146
10000	0.0002	0.0002	0.00013	0.00011	0.0115	0.0104

Table 2: Coverage of the nominal 95% confidence interval and rejection frequency of Hansen's J test, by sample size.

n	Coverage (1-step)	Coverage (2-step)	J-test rejection
50	0.918	0.914	0.053
100	0.933	0.927	0.050
250	0.941	0.938	0.052
500	0.950	0.947	0.042
1000	0.951	0.952	0.050
2500	0.951	0.950	0.052
5000	0.949	0.946	0.048
10000	0.945	0.948	0.048

5 Discussion and Conclusion

The theory of Section 2 and the simulation of Sections 3–4 tell the same story from two directions. Propositions 1–4 are exact limiting results, and the Monte Carlo evidence confirms every one of them is eventually a very good approximation: bias and variance shrink toward zero, the two-step estimator dominates the one-step estimator in variance exactly as the efficiency bound (7) predicts, the studentized estimator's distribution converges visibly to standard normal, confidence-interval coverage approaches its nominal level, and the J test is correctly sized. None of this is automatic at the sample sizes an applied researcher actually has: at $n = 50$, coverage is several percentage points below nominal for both estimators, and the studentized distribution is visibly non-normal, even though the design avoids the weak-instrument pathologies that make GMM/IV inference unreliable for reasons unrelated to sample size (Section 2.7, caveat 3). The practical implication is not that GMM inference is unreliable in general, but that the gap between "consistent and asymptotically normal" and "trustworthy at my sample size" has to be checked, not assumed – exactly the exercise this handout carries out.

The companion script, `scripts/gmm_finite_sample_asymptotic_sim.R`, is the single computational source for every number and figure in this handout: it implements the closed-form estimator (3), the sandwich variances (6) and (8), and Hansen's J statistic exactly as derived in Section 2, and can be re-run (`Rscript scripts/gmm_finite_sample_asymptotic_sim.R`) to reproduce every result reported here, or edited to explore other sample-size grids, degrees of heteroskedasticity, or instrument strength.

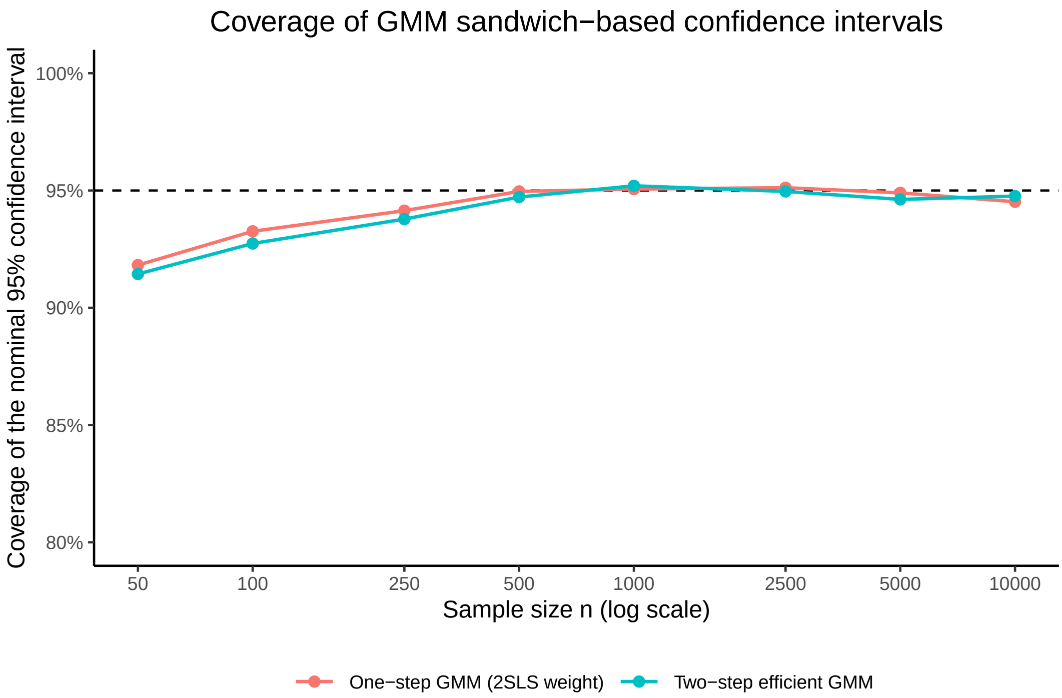

Figure 5: Empirical coverage of the nominal 95% confidence interval for the GMM slope estimators, using the sandwich standard errors of equations (6) and (8).

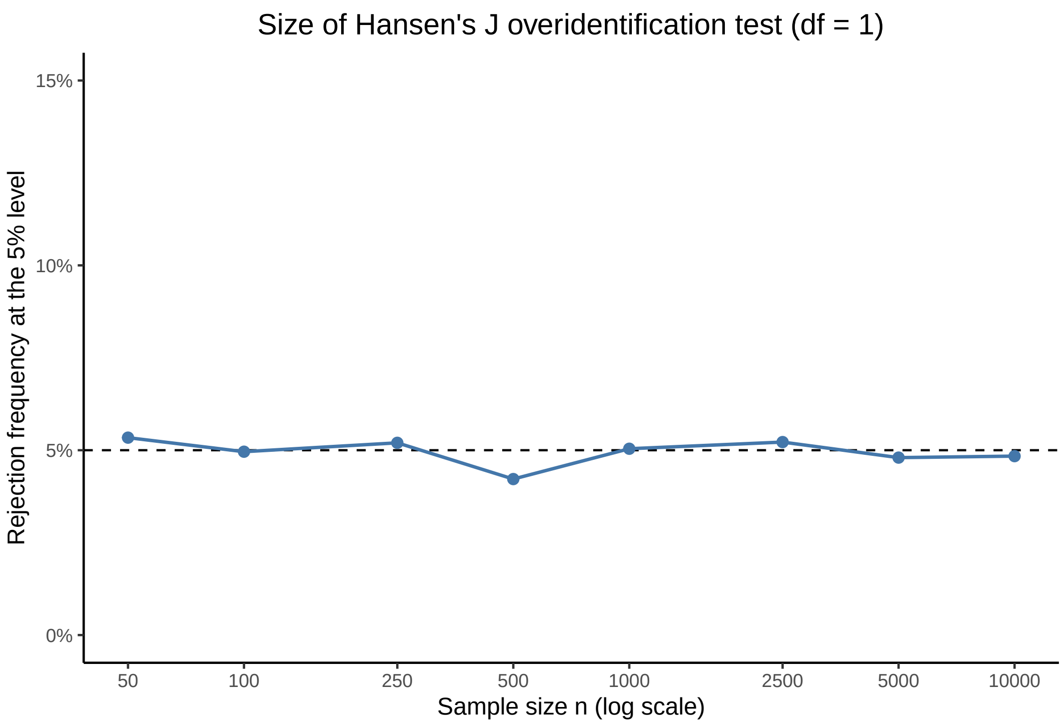

Figure 6: Rejection frequency of Hansen's J overidentification test (df = 1) at the nominal 5% level, under correct specification.

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